

Mockito

Exercise 1: Mocking and Stubbing

Problem

Mock an external API and stub its method.

Setup

Create ExternalApi, MyService and MyServiceTest.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);
when(api.getData()).thenReturn("Mock Data");
MyService s=new MyService(api);
assertEquals("Mock Data",s.fetchData());
```

Output

Program executes successfully and the Mockito test passes.

Exercise 2: Verifying Interactions

Problem

Verify that a mocked method is called.

Setup

Create a mock object and verify interaction.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);
MyService s=new MyService(api);
s.fetchData();
verify(api).getData();
```

Output

Program executes successfully and the Mockito test passes.

Exercise 3: Argument Matching

Problem

Verify method calls using argument matchers.

Setup

Use anyString() matcher.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);  
api.sendData("Hello");  
verify(api).sendData(anyString());
```

Output

Program executes successfully and the Mockito test passes.

Exercise 4: Handling Void Methods

Problem

Test a void method using Mockito.

Setup

Stub a void method using `doNothing()`.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);  
doNothing().when(api).clearData();  
api.clearData();  
verify(api).clearData();
```

Output

Program executes successfully and the Mockito test passes.

Exercise 5: Multiple Returns

Problem

Stub a method to return different values.

Setup

Use `thenReturn()` with multiple values.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);  
when(api.getData()).thenReturn("One","Two");  
System.out.println(api.getData());  
System.out.println(api.getData());
```

Output

Program executes successfully and the Mockito test passes.

Exercise 6: Verifying Interaction Order

Problem

Verify methods are called in sequence.

Setup

Use InOrder.

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);  
api.connect();  
api.disconnect();  
InOrder order=inOrder(api);  
order.verify(api).connect();  
order.verify(api).disconnect();
```

Output

Program executes successfully and the Mockito test passes.

Exercise 7: Void Methods with Exceptions

Problem

Stub a void method to throw an exception.

Setup

Use doThrow().

Java Implementation

```
ExternalApi api=mock(ExternalApi.class);  
doThrow(new RuntimeException()).when(api).clearData();  
assertThrows(RuntimeException.class,  
()->api.clearData());
```

Output

Program executes successfully and the Mockito test passes.